

PAPER395 — Wormhole UQFF Acceleration: 13th Resonance Term $a_{\text{worm}} = f_{\text{worm}} \cdot E_{\text{vac_neb}} / (b^2 + r^2)$

Session: 107 (groksharecfdcad2f5.txt deep re-analysis pass)

PAPER395 — Wormhole UQFF Acceleration: 13th Resonance Term $a_{\text{worm}} = f_{\text{worm}} \cdot E_{\text{vac_neb}} / (b^2 + r^2)$

Source: groksharecfdcad2f5.txt, lines ~900–1300 (C++ unit tests + MUGE.h) **Section:** test_compute_a_wormhole() unit test and compute_a_wormhole() in MUGE.cpp **Session:** 107 (groksharecfdcad2f5.txt deep re-analysis pass) **CP4** **Class:** WormholeUQFFResonanceAccelerationCalculator (CP4 #46)

1. Overview

PAPER373 (Morris-Thorne Wormhole Null Geodesics) covered wormhole topology and geodesic paths through the UQFF lens. PAPER395 introduces a **distinct formula**: the wormhole contribution as an **acceleration term** (the 13th resonance term in the MUGE Resonance equation). This is not the geodesic path but the effective UQFF acceleration generated by the wormhole throat geometry.

The formula appears in the C++ compute_a_wormhole() function and its unit test, both explicitly in the groksharecfdcad2f5 thread.

2. The Wormhole Acceleration Formula

2.1 Master Equation

$$a_{\text{worm}} = \frac{f_{\text{worm}} \cdot E_{\text{vac_neb}}}{b^2 + r^2}$$

2.2 Parameter Definitions

Parameter	Value	Physical Meaning
f_{worm}	1.0 (default)	Wormhole topology coupling factor
$E_{\text{vac_neb}}$	$7.09 \times 10^{-36} \text{ J/m}^3$	Nebular vacuum energy density
b	1.0 m (default)	Wormhole throat radius
r	distance from throat (m)	Observer radial distance

2.3 Limiting Forms

Near the throat ($r \ll b$):

$$a_{\text{worm}} \approx \frac{E_{\text{vac_neb}}}{b^2} = \frac{7.09 \times 10^{-36}}{1.0} = 7.09 \times 10^{-36} \text{ m/s}^2$$

Far from the throat ($r \gg b$):

$$a_{\text{worm}} \approx \frac{E_{\text{vac_neb}}}{r^2}$$

This has the **inverse-square fall-off** typical of gravitational acceleration.

3. Unit Test Verification

From `test_compute_a_wormhole()` in `UnitTests.cpp`:

```
void test_compute_a_wormhole() {
    double r = 1e4; // 10 km from throat
    double b = 1.0; // 1 m throat radius
    double expected = 7.09e-36 / (1.0 + r * r);
    // expected = 7.09e-36 / (1 + 1e8) = 7.09e-36 / 1.000000001e8 ≈ 7.09e-44 m/s2
    double result = compute_a_wormhole(r);
    assert(std::abs((result - expected) / expected) < 1e-6);
}
```

Test values:

$r = 10^4 \text{ m}$ (10 km), $b = 1 \text{ m}$

$a_{\text{worm}} = 7.09 \times 10^{-36} / (1 + 10^8) \approx 7.09 \times 10^{-44} \text{ m/s}^2$

This is an **extremely small acceleration**, physically consistent with the sub-Planck scale of wormhole UQFF effects at astrophysical distances.

4. Role in Resonance MUGE

4.1 Position in the 13-Term Resonance Sum

The MUGE Resonance equation is a sum of 13 independent acceleration terms:

```
g_{\text{res}} = a_{\text{DPM}} + a_{\text{THz}} + a_{\text{vac,diff}} + a_{\text{super}}
+ a_{\text{Aether,res}} + U_{\text{g4i}} + a_{\text{quantum}} + a_{\text{Aether,freq}} +
a_{\text{fluid,freq}} + a_{\text{osc}} + a_{\text{exp,freq}} + f_{\text{TRZ}} +
a_{\text{worm}}
```

a_{worm} is the **13th and final term**, representing the wormhole backreaction on local spacetime curvature.

4.2 Magnitude Comparison

For the canonical 7-system evaluation at $r = 10^{12} \text{ m}$ (SgrA*):

```
a_{\text{worm}} = \frac{7.09 \times 10^{-36}}{1 + (10^{12})^2} =
\frac{7.09 \times 10^{-36}}{10^{24}} \approx 7.09 \times 10^{-60} \text{ m/s}^2
```

Compared to $a_{\text{DPM}} \sim 10^{100} \text{ m/s}^2$ (SgrA* resonance dominant), the wormhole term contributes negligibly to the resonance sum for compact objects but could dominate near the wormhole throat itself.

5. Connection to Morris-Thorne Geometry

The Morris-Thorne wormhole metric is (PAPER_373):

```
ds^2 = -e^{2\Phi(r)}c^2dt^2 + \frac{dr^2}{1-b(r)/r} + r^2 d\Omega^2
```

The UQFF wormhole acceleration formula can be derived from the **vacuum energy gradient** near the throat:

$$a = -\nabla\left(\frac{E_{\text{vac}}}{b^2 + r^2}\right) \sim \frac{2r}{b^2 + r^2} \cdot E_{\text{vac}}$$

The formula $E_{\text{vac,neb}}/(b^2+r^2)$ represents the **potential** rather than the gradient, but is used as a proxy acceleration in the resonance MUGE framework (consistent with how all resonance terms are treated as effective acceleration contributions).

6. Comparison to Existing Papers

Paper	Formula	Distinction
PAPER_373	Morris-Thorne geodesics, exotic matter	Topology / null geodesics
PAPER_375	Wormhole + Meissner relativistic γ	Lorentz + Meissner blend
PAPER_377	Wormhole safety check	Stability bounds
PAPER_395	$a_{\text{worm}} = f_w \cdot E_{\text{vac}}/(b^2+r^2)$	13th resonance term acceleration

7. C++ Implementation

```
double compute_a_wormhole(double r, double b = 1.0, double f_worm = 1.0,
double Evac_neb = 7.09e-36) {
return f_worm * Evac_neb / (b * b + r * r);
}
// Default: b=1.0 m throat, f_worm=1.0, Evac_neb=7.09e-36 J/m³
// At r=1e4: a_worm = 7.09e-36 / (1 + 1e8) ≈ 7.09e-44 m/s²
// At r→0: a_worm → 7.09e-36 m/s² (throat value)
```

8. Physical Context

8.1 $E_{\text{vac,neb}} = 7.09 \times 10^{-36} \text{ J/m}^3$

This vacuum energy density value appears consistently throughout the UQFF resonance equations (also used in a_{DPM} , $a_{\text{vac,diff}}$, a_{Ug4i}). It represents the **nebular vacuum floor** — the minimum vacuum energy density in star-forming nebula environments (Pillars of Creation, Tapestry of Blazing Starbirth, etc.).

8.2 Throat Radius $b = 1.0 \text{ m}$

The default throat radius of 1 meter gives a **macroscopic but sub-stellar** scale. In the UQFF framework, this corresponds to a hypothetical stellar-interior wormhole where the throat is threaded by Aether strings of density $\rho_A = 10^{-23} \text{ kg/m}^3$.

9. Summary

PAPER395 formalizes $a_{\text{worm}} = f_{\text{worm}} \cdot E_{\text{vac,neb}} / (b^2 + r^2)$ as the **13th resonance MUGE term**. With default parameters ($b=1 \text{ m}$, $E_{\text{vac}}=7.09 \times 10^{-36} \text{ J/m}^3$), the throat acceleration is $7.09 \times 10^{-36} \text{ m/s}^2$ and falls as r^{-2} at large distances. Unit test confirms value $7.09 \times 10^{-44} \text{ m/s}^2$ at $r=10^4 \text{ m}$

with tolerance $<10^{-6}$. This term completes the 13-term resonance MUGE summation alongside PAPER381 terms.